

Evaluating Credit Card Minimum Payment Restrictions

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The views presented are those of the authors and do not necessarily reflect those of the Bank of Canada

Research Question

What is the magnitude of the trade-offs of a hard paternalistic policy that restricts consumer choices?

Contributions

- Behavioral Public Economics:
 - Evaluating hard paternalism
- Credit Cards & Consumer Financial Protection Policy:
 - Equilibrium responses to a market-wide credit card policy

1. Motivation
2. Quebec Policy
3. Data & Methodology
4. Effects on Consumers
5. Supply Responses

Motivation

Stylized Motivating Example: Why ↓ Credit Card Debt?

Imagine two types of consumers use credit cards: (Heidheus & Köszegi, 10, 15; Campbell, 16)

- “**Sophisticates**” optimize,
- “**Naïves**” overborrow (e.g., financially unsophisticated, naïve present focus, targeting)
- Both types experience binding **liquidity constraints**

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 - b) Avoid constraining sophisticates' choices (↑ minimum payment, ↓ credit card access)

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Unfortunately “ideal policies” of information & nudges ineffective (e.g., Agarwal et al. 15; Keys & Wang, 15; Seira et al., 17; Adams et al. 22; Batista et al., 24; Guttman-Kenney et al., 25)
→ “nudges are not enough” (Laibson, 20)

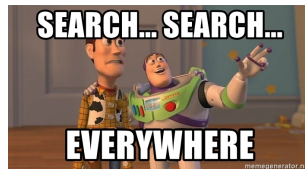
Need to evaluate hard paternalistic policies trade-off benefits of 1 vs. costs of 2

Supply responses may further complicate what is “ideal” policy

Hard Paternalistic Policy To ↓ Credit Card Debt?

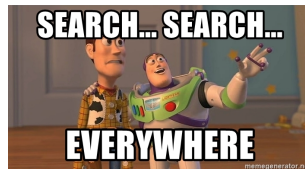
Mandating tighter (higher) credit card minimum payments

- Want policy forcing credit card lenders involuntarily ↑ minimum payment



Where to Study Tighter Minimum Payments Policy

- Want policy forcing credit card lenders involuntarily $\uparrow$ minimum payment
- Policy variation does not exist in U.S.
- Prior research studied lenders *voluntarily* changing minimum payments (D'Astous & Shore, 17; Keys & Wang, 19, Castellanos et al., 24; Medina & Negrin, 22), & Turkish policy effectively preventing using credit cards for borrowing (Agarwal et al., 23)
- Takes us to Quebec, Canada!



Quebec Policy

Quebec: Laboratory for Hard Paternalism Aiming to ↓ Credit Card Debt

Quebec office de la protection du consommateur:

*“These changes are part of the measures adopted by the government
to prevent indebtedness”*

**“It is advantageous to pay the balance of your credit card each month,
*because no credit charges are then applicable”***

Quebec rolls out new credit card rules aimed at lowering high household debt

By Reuters Staff

2 MIN READ



OTTAWA, Aug 2 (Reuters) - The Canadian province of Quebec has become the first in the country to require minimum credit card payments as legislators try to tackle rising household debt levels.

Quebec introduces credit card regulations aimed at curbing consumer debt

CTVNews.ca Staff



Ben Cousins CTVNews.ca Writer
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Published Thursday, August 1, 2019 4:38PM EDT

No credit card minimum payment requirements:

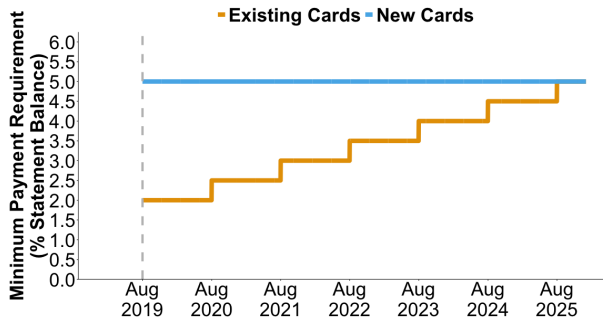
- Pre-August 2019: All Canadian provinces
- Post-August 2019: All Canadian provinces except Quebec

Quebec Policy Experiments

No credit card minimum payment requirements:

- Pre-August 2019: All Canadian provinces
- Post-August 2019: All Canadian provinces except Quebec

Quebec introduced credit card minimum payment policy from August 2019:



Data & Methodology

Credit Reporting Data

Bank of Canada's anonymized TransUnion consumer credit reporting data

- Monthly, tradeline-level information on each credit card held in Canada
- Includes account dates, balances, credit limits, minimum payments, actual payments
- January 2018 to November 2024

(See Gibbs, Guttman-Kenney, Lee, Nelson, Van der Klaauw & Wang, 2025 *JEL* guidance!)

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Mintel CompereMedia Data

- Credit card offers received via mail/email by 8,000 Canadian households
- Includes contract terms
- Measure of supply (e.g., Ru & Schoar, 2020; Grodzicki, 2022, Han, Keys, & Li 2018)

Synthetic Difference-in-Differences (SDID) Methodology

Arkhangelsky, Athey, Hirshberg, Imbens & Wager (2021 AER); Arkhangelsky & Imbens (2024)
- SDID outperforms both synthetic controls & difference-in-differences approaches.

Follow Arkhangelsky et al. (2021) estimating:

$$(\hat{\delta}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\delta, \mu, \alpha, \beta}{\operatorname{argmin}} \left\{ \sum_{p=1}^9 \sum_{t=-18}^6 (y_{p,t} - \mu - \alpha_p - \beta_t - \delta(\mathbb{I}\{t \geq 0\}_t \times \mathbb{I}\{QUEBEC\}_p))^2 \hat{\omega}_p \hat{\lambda}_t \right\}. \quad (1)$$

- Outcomes in Quebec (Q) are compared to “Synthetic Quebec” (SQ) of 8 provinces
- Weights using statement balances $t = -18$ to $t = -13$ to allow for anticipation effects

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- Outcomes in Quebec (Q) are compared to “Synthetic Quebec” (SQ) of 8 provinces
- Weights using statement balances $t = -18$ to $t = -13$ to allow for anticipation effects
- Same weights for all outcomes (weights range from 0.11 to 0.14)
- Standard errors using placebo variance estimation to account for single treated cluster
- Show dynamic effects relative to pre-policy baseline ($t = -18$ to $t = -13$):

$$\delta_{\tau} = \left(\bar{y}_t^Q - \bar{y}_t^{SQ} \right) - \left(\bar{y}_{baseline}^Q - \bar{y}_{baseline}^{SQ} \right) \quad (2)$$

Effects on Consumers

How Lenders Implemented? **Non-binding**, **Bind**, Bind and go beyond

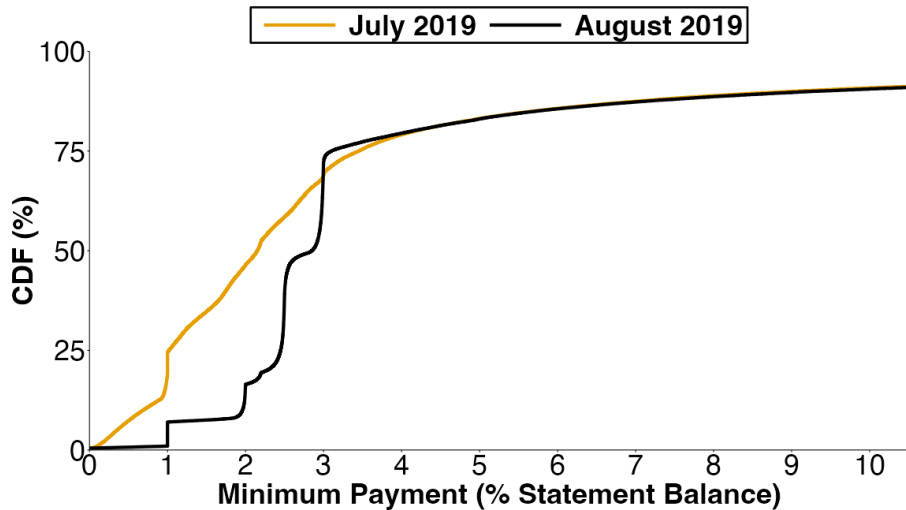
Lender	July 2019		August 2019	
	Non-Quebec	Quebec	Non-Quebec	Quebec
A	\$10 + interest + fees		max {2.0% statement balance, \$10}	
B,C,D	\$10 + interest + fees		max {2.5% statement balance, \$10}	
E	\$10 + interest + fees		max {3.0% statement balance, \$10}	
F	max {1.0% statement balance, \$10}		max {2.0% statement balance, \$10}	
G	max {1.0% statement balance, \$10}		max {3.0% statement balance, \$10}	
H	max {2.0% statement balance, \$10}		max {2.0% statement balance, \$10}	
I	max {2.5% statement balance, \$10}		max {2.5% statement balance, \$10}	
J	max {3.0% statement balance, \$10}		max {3.0% statement balance, \$10}	

After 2019, lenders typically $\uparrow$ 0.5 percentage points each year until reaches 5%

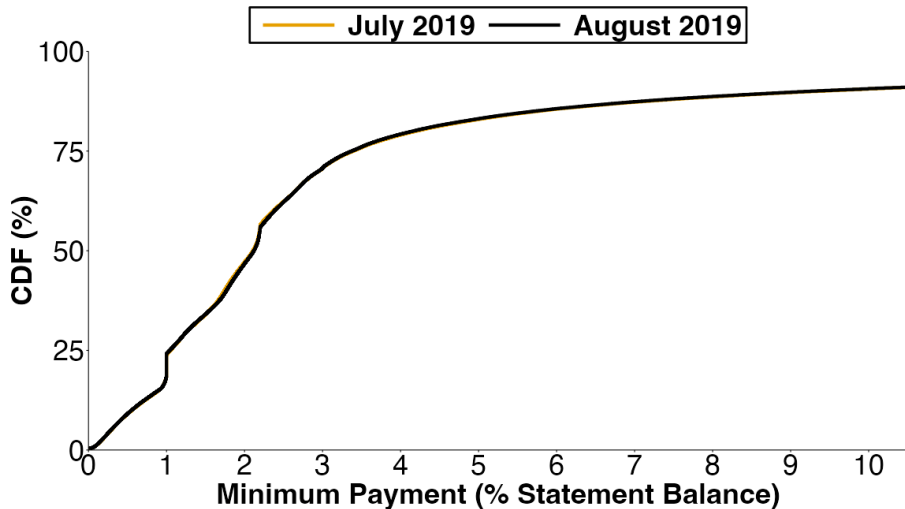
New cards opened from August 2019: all lenders set minimum payments as:

max {5.0% statement balance, \$10}

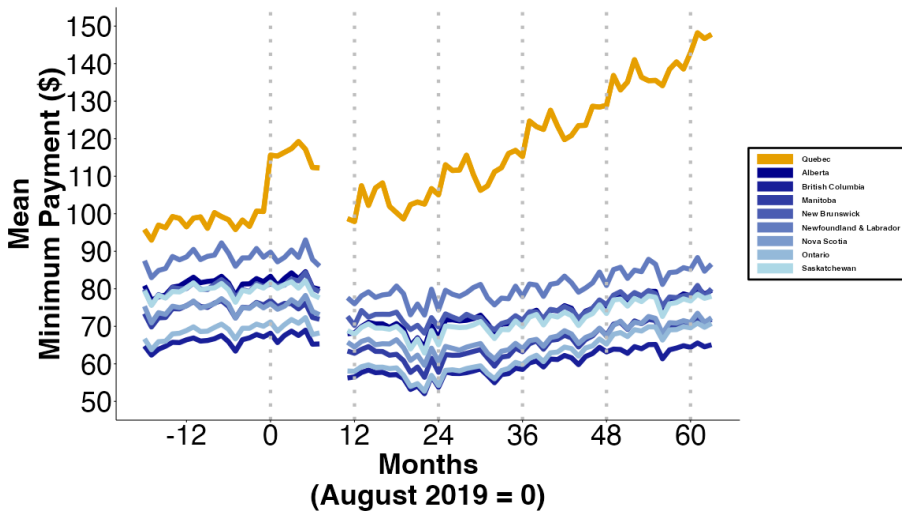
↑ Minimum Payments in Quebec



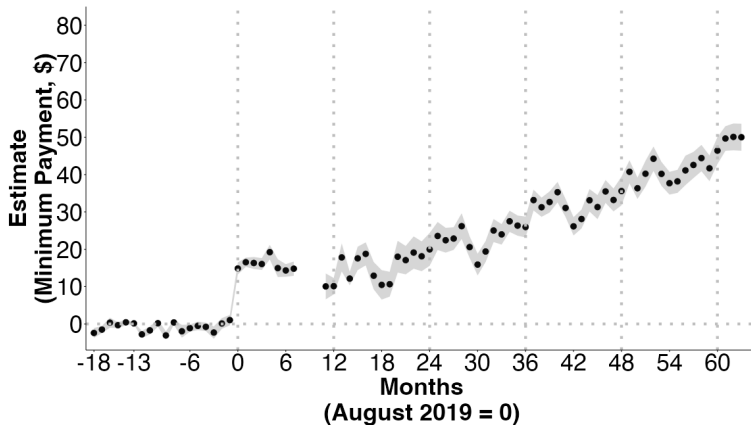
No Change in Minimum Payments in Ontario (or Other Provinces)



Unconditional Means Show $\uparrow$ Minimum Payments in Quebec, No Increase in Other Provinces



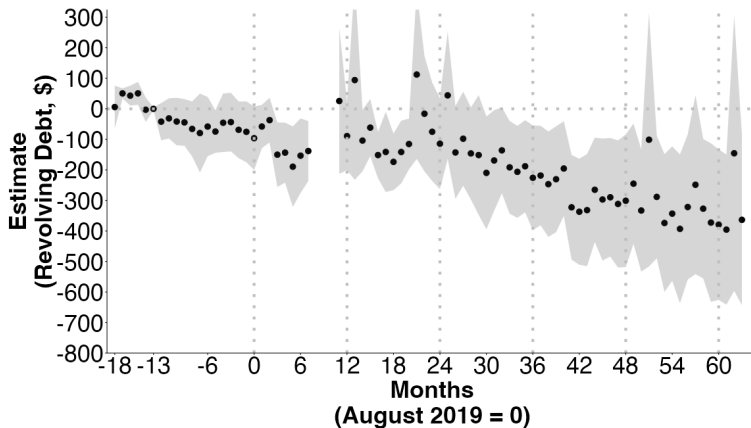
SDID 'First Stage': $\uparrow$ Minimum Payments in Quebec



$t+1$: $\uparrow$ \$16.5 (s.e. \$0.6), $t+6$: $\uparrow$ \$14.3 (s.e. \$0.9), $t+60$: \$46.4 (s.e. \$1.8)

Baseline mean: \$98

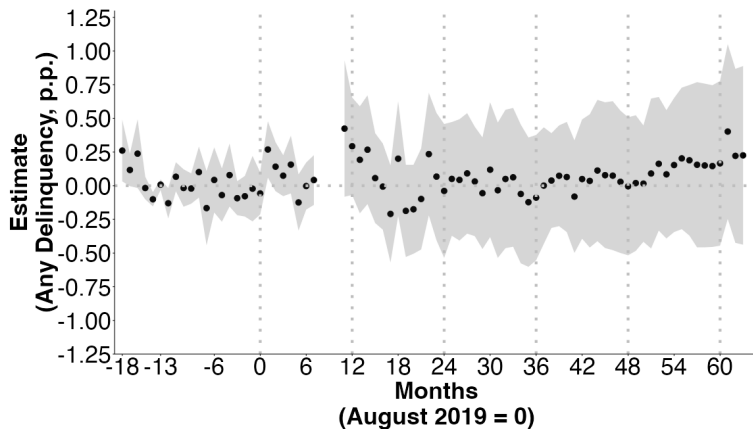
SDID: ↓ Credit Card Revolving Debt (Statement Balance - Payments)



t+6: ↓ \$182.8 (s.e. \$61.4), t+60: ↓ \$379.3 (s.e. \$125.7)

Baseline mean: \$1,240

SDID: Effects on Any Credit Card Delinquency (Any 30+ Days Past Due)

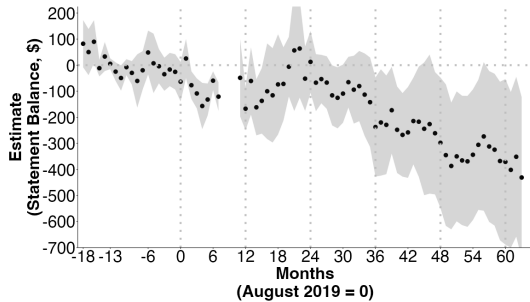


t+1: $\uparrow$ 0.2689 percentage points (s.e. 0.1066), t+6: $\downarrow$ 0.0025 (s.e. 0.0890)

Baseline mean: 2.6891 percentage points

Statement Balances ↓ & Credit Limits ↓

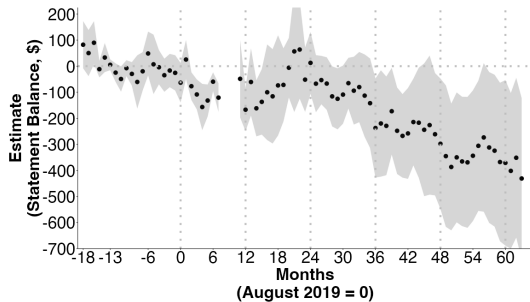
Statement Balances



Baseline mean: \$3,128

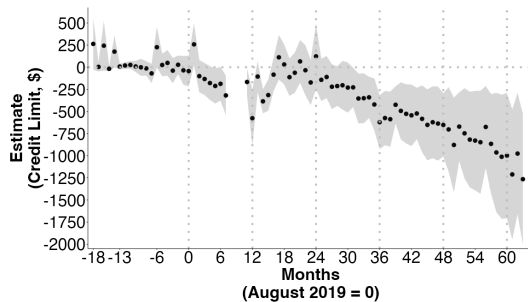
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Statement Balances



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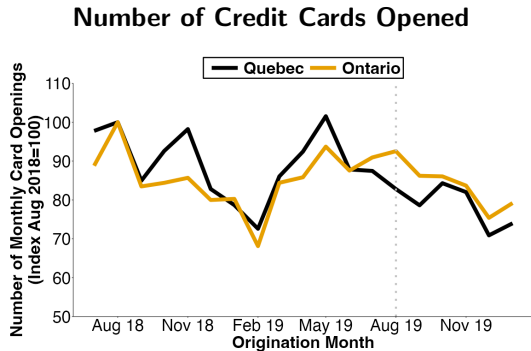
Credit Limits



Baseline mean: \$12,299

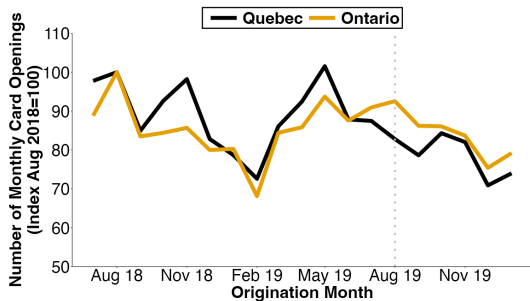
Supply Responses

↓ New Credit Cards Opened, and ↓ Value of New Credit Card Limits

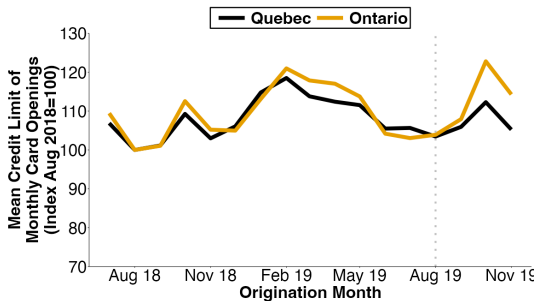


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Number of Credit Cards Opened

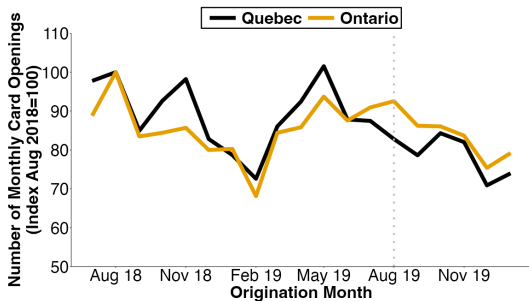


Mean Credit Limits of Cards Opened

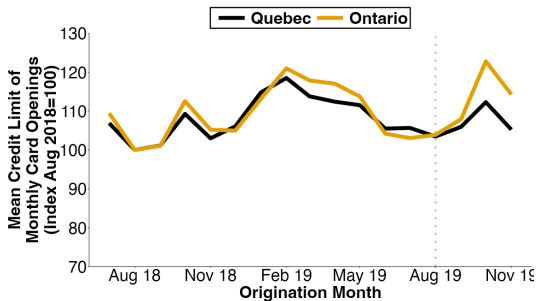


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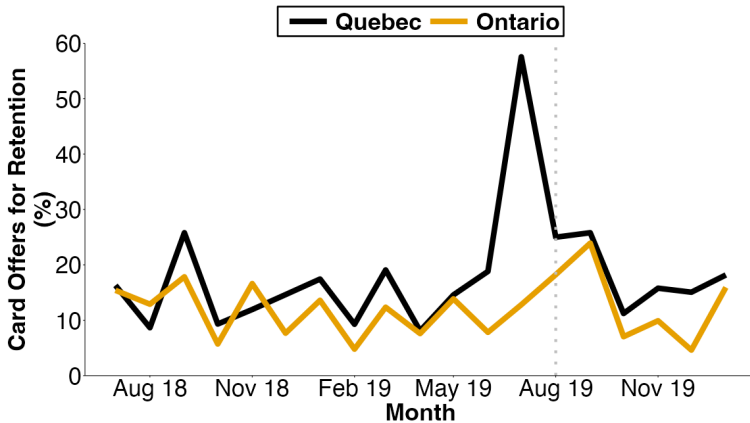


Mean Credit Limits of Cards Opened



Consistent with higher minimum payments making some consumers unprofitable

Mintel Data Shows Spike in Mailed Offers to Retain Existing Cardholders



Card offers in Quebec just before August 2019 appear to have
↑ credit limits, ↓ APRs, and ↓ annual fees

Conclusions

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- Use comprehensive Canadian credit reporting data and a synthetic difference-in-differences design to estimate the causal effects of a hard paternalistic policy
- Quebec policy significantly & permanently $\uparrow$ minimum payments, permanently $\downarrow$ debt, and only temporarily $\uparrow$ delinquency
- Policy significantly & permanently $\downarrow$ credit access

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Next steps

- Heterogeneity by (i) minimum payment exposure, (ii) behavioral consumers
- Spillovers e.g., other debt balances? delinquency? credit scores, autos & mortgages
- Are improved behaviors on new cards with 5% minimum payment due to selection?
- Who loses credit access? revolvers (cross-subsidization test)? what happens to them?

Thank you!

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